

Research Article

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Extracting Inductive Bias Using Generative Models

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Abstract: Inductive biases play a central role in the generalization and data efficiency of neural networks. Network connectivity encodes such biases, yet its design often relies on domain expertise. This paper presents a generative approach to discovering connectivity patterns shared across related tasks. For each task, a separate generative model captures a family of connectivity patterns extracted from trained networks. These generative models are used to refine the inductive bias shared across tasks. The central hypothesis is that agreement across task-specific generative models reveals structural inductive biases that generalize to previously unseen tasks.

Keywords: inductive bias, generative model

1 Introduction

test test

References

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